

Showcasing MobileGravity

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Figure 1: Procedure for adjusting the perceived weight through MobileGravity. The four images on the left show connecting MobileGravity's handheld proxy object with its base station, adjusting its weight, and disconnecting both components. The user can then move around freely (image on the right). The outline of the liquid reservoir inside the proxy object is highlighted.

ABSTRACT

Providing weight sensations in Virtual Reality (VR) is a persistent challenge. In this video, we present MobileGravity [3], a novel mobile weight interface utilizing liquid transfer to change the weight of a handheld VR proxy. MobileGravity overcomes the conflict between mobility and performance by decoupling its heavy components, such as the actuator and the liquid supply, from the weight-changing object. This enables the system to apply weight changes of up to 1 kg in 235 g/s and yet allows users to walk around and turn in any direction. The video showcases the interaction with MobileGravity, sheds light on its technical implementation, and illustrates previously restricted use cases.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices; Virtual reality.**

KEYWORDS

virtual reality, haptics, weight perception, weight simulation

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CONCEPT

Perceiving the weight of objects is important to understand their properties and to interact with them naturally [1]. Sensing weight in VR however, is restricted due to the static, low weight of commercial VR controllers and limitations of research prototypes regarding accuracy, mobility, comfort, and the conveyance of heavy weights [6]. Utilizing liquid transfer to change the weight of handheld or body-worn devices has emerged as a promising approach [4, 5, 7]. Yet, previous designs do not allow for lightweight and mobile prototypes without compromising on speed, complexity, and range of weight that can be rendered [2].

We present MobileGravity [3], a system, that decouples the weight-changing object from the liquid supply and the actuator. This enables the incorporation of powerful actuators and a high-volume water tank while permitting free movement. MobileGravity is therefore particularly valuable to scenarios that require effective locomotion or unhindered movement. It could, for instance, be used to convey object weight in training and education scenarios, to sense heavy and lightweight items in VR games, to simulate avatar weight, or to change intensities of VR workouts.

MobileGravity consists of a mobile handheld proxy object and a base station that can pump up to 1 l of water into or out of the proxy object at a rate of 235 g/s. The proxy object comprises a 1 l water bag, two handles and an HTC Vive Tracker. The base station consists of a bi-directional pump, a 10 l water tank, two solenoid actuators, and a limit switch to detect whether a physical connection to the proxy object is established. An Arduino Micro controls the components of the base station using four relay modules.

To connect and disconnect the proxy object and the base station when the weight needs to be adjusted, MobileGravity employs Koolance QD3 Quick Disconnect No-Spill Couplings, which are fittings that automatically cut off the water flow when disconnected. The male fitting² is integrated into the proxy object, while the corresponding female fitting³ is embedded in the base station. Users can insert the proxy object into a recess at the top of the base station. This establishes the connection between both units and starts the weight adjustment. To maintain immersion, this insertion is carried out through the user placing a virtual object at a specific location in the virtual environment. When the weight adjustment is completed, the solenoid actuators pull down a safety ring to release the proxy object from the base station.

To allow users to experience different centers of mass, the proxy object can be grasped at two distinct handles. The extra handle is orthogonal to the primary handle. This further increases the variety of objects that can be simulated, as it enables users to perceive the center of weight either at the top, bottom, front, back, left, or right of their hand, depending on the handle and grip angle when picking up the proxy object.

For users to be able to interact with the virtual world while the proxy object is in the base station, their hands are continuously tracked and displayed in the virtual world. Cameras built into or attached to the VR headset, such as the Leap Motion Controller, facilitate the hand tracking.

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²<https://koolance.com/qd3-ms13x19-bk-quick-disconnect-no-spill-coupling-male-for-13mm-x-19mm-1-2in-x-3-4in-black>

³<https://koolance.com/qd3-fs13x19-bk-quick-disconnect-no-spill-coupling-female-for-13mm-x-19mm-1-2in-x-3-4in-black>